

Ethical Hacking Viva Master Guide (Experiments 1–15)

Experiment-wise concepts, commands, definitions, and likely viva questions.

Exp 1 Metadata2Go

- What is metadata? Data about data.
- Why important? Helps identify author, dates, software and source of files.
- What did we do? Upload file and inspect metadata.
- Output: Author, created date, modified date, software used.

Exp 2 FTK Imager

- What is FTK Imager? Forensic acquisition and evidence viewing tool.
- What did we do? Added evidence folder and viewed files.
- What is digital evidence? Any electronic data useful in investigation.
- Why use FTK? Read-only evidence analysis.

Exp 3 Autopsy

- What is Autopsy? Open-source forensic investigation platform.
- What did we do? Create case, add logical files, run analysis.
- What is hash lookup? Identify known files by hashes.
- What is file type identification? Detect actual file type regardless of extension.

Exp 4 Nessus

- What is vulnerability assessment? Finding security weaknesses.
- What did we do? Scan localhost using Basic Network Scan.
- Difference between VA and Pentest? VA finds flaws; Pentest exploits them.
- What is CVE? Common Vulnerabilities and Exposures.

Exp 5 John the Ripper

- What is password cracking? Recovering passwords from hashes.
- Commands used: zip2john extracts hash, john cracks it.
- What is a dictionary attack? Uses predefined password lists.
- Difference between dictionary and brute force? Brute force tries all combinations.

Exp 6 Hydra

- What is Hydra? Online login cracker.

- Why start SSH first? Hydra attacks a live service.
- Meaning of -l? Username.
- Meaning of -P? Password file.
- Difference between Hydra and John? Hydra attacks services, John attacks hashes.

Exp 7 Metasploit

- What is privilege escalation? User gains admin/root access.
- Vertical escalation? User to root/admin.
- Horizontal escalation? User A to User B.
- What is Meterpreter? Advanced post-exploitation shell.
- What did local_exploit_suggester do? Suggest escalation opportunities.

Exp 8 Steghide

- What is steganography? Hiding data inside another file.
- Difference between encryption and steganography? Encryption hides content, steganography hides existence.
- Cover file? Original image.
- Embedded file? Hidden secret file.
- Commands: embed and extract.

Exp 9 SET Toolkit

- What is social engineering? Manipulating people for information.
- What is phishing? Fake emails/websites to steal credentials.
- What is credential harvesting? Collecting usernames and passwords.
- What did we do? Mass Mailer Attack menu flow.

Exp 10 Nmap Stealth Scan

- What is Nmap? Network Mapper.
- What is SYN scan (-sS)? Half-open stealth scan.
- What is TCP scan (-sT)? Full TCP connection.
- What is -f? Packet fragmentation.
- What is -D RND:5? Decoy scan using fake IPs.

Exp 11 Session Hijacking

- What is a session? Authenticated connection between client and server.
- What is a session cookie? Identifier maintaining login state.
- What is session hijacking? Stealing session tokens.
- Why HTTPS? Protects cookies in transit.

Exp 12 DoS/DDoS

- DoS? Single attacker overwhelms target.
- DDoS? Multiple attackers overwhelm target.
- What is a botnet? Collection of compromised devices.
- What did Wireshark show? Large traffic spikes and packet patterns.

Exp 13 Shodan/Nmap

- What is Shodan? Search engine for internet-connected devices.
- What is -sn? Host discovery.
- What is -sV? Service detection.
- What is -O? OS detection.
- What is -A? Aggressive scan.

Exp 14 OpenSSL/GPG

- What is encryption? Convert plaintext to ciphertext.
- What is decryption? Recover plaintext.
- What is AES-256? Symmetric encryption using 256-bit key.
- What did OpenSSL do? Encrypt and decrypt files.
- What did GPG do? Password-protected file encryption.

Exp 15 Hashcat

- What is hashing? One-way conversion of data.
- Can hashes be reversed? No, only cracked by guessing.
- What is MD5? 128-bit hashing algorithm.
- Meaning of -m 0? MD5 mode.
- Meaning of -a 0? Dictionary attack mode.
- What did we recover? Password 'hello' from its MD5 hash.